

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

20/20
I, V. Sheersika (192472056) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Sheersika

Problem Identification

1. configured IP address 192.168.10.65 belongs to Subnet 192.168.10.64/26
2. default gateway 192.168.10.1 belongs to Subnet 192.168.10.0/26.
3. Since workstation & gateway are in diff. Subnets.

Debugging

Calculate subnet: Subnet mask = /26

Block size = 64

Subnet size = 64

Subnet

192.168.10.0/26

192.168.10.64/26

192.168.10.128/26

192.168.10.192/26

Address Range

0 - 63

64 - 127

128 - 191

192 - 255

Optimization

Assign both the workstation & gateway to the same Subnet

Device

Gateway

workstation

correct address

192.168.10.65

192.168.10.66

Analysis

1. Communication failure occurred because the gateway belonged to different subnet

5. Final optimized configuration

network address - 192.168.10.64

Broadcast address - 192.168.10.127

Host range - 192.168.10.65 - 192.168.10.126

Gateway - 192.168.10.65

workstation - 192.168.10.66

2. Problem Identification:

Given

Device A: 2001::db8::ff00:42:8329/64

Device B: 2001::db8::0:1:1:1/64

Debugging:

expanded express

Device A:

2001:0db8:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix
with 164

Device A

2001:0db8:0000:0000::/64

Device B

2001:0db8:0000:0000::/64

Both belong to same subnet

Problem:

no prefix mismatch

communication failure may due to

1. Firewall

2. Router configuration.

Optimization

Example addresses

Device A:

2001:db8::10/64

Device B:

2001:db8::20/64

Analysis:

Both devices already share the same network prefix. IPv6 addressing is correct. network troubleshooting should focus.

3. Problem Identification:

Given:

192.168.1.0/24

Departments:

1. 126

2. 127

3. 128.

Some department have excess addresses.

Debugging

1. 126

Network - 192.168.1.0

Host range - 192.168.1.1 - 192.168.1.62

Broad Cast - 192.168.1.63

usable hosts - 62

2. 127

network - 192.168.1.64

Host range - 192.168.1.65 - 192.168.1.94

Broad Cast - 192.168.1.95

usable hosts - 30.

198

Network - 192.168.1.96

Host range - 192.168.1.97 - 192.168.1.11

Broad Cast - 192.168.1.111

Usable Hosts - 14

Analysis

Department need only 10 hosts
allocated 126

unused = 62 - 10

= 52 hosts wasted

4. Expanded addresses

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix:

2001:0db8:0000:0000:0001:0000:0000:0001

Optimized address

Device A:

2001:db8::10/64

Device B:

2001:db8::20/64

available Hosts = 18,446,774,073,709,551,616

available Hosts:

$$2^{64} = 18,446,744,073,709,571,616$$

5. Problem Identification:

Destination

next hop

192.168.1.0/24

10.0.0.2

192.168.2.0/24

10.0.0.3

default

10.0.0.1

Destination IP:

192.168.2.100

Debugging:

Destination belongs to 192.168.2.0/24

network Range: 192.168.2.1 - 192.168.2.254

matching route: 192.168.2.0/24

longest prefix match

Router selects: 192.168.2.0/24

next hop: 10.0.0.3

Verify:

Destination IP:

192.168.2.100 is valid.

Q. J. J.